

Week 9 – System Requirements Document

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Week: Week 9 – Designing & Prototyping Human-Centred Systems (HCI/HRI)

Chosen Implementation: AI-Assisted Emergency Department Triage Support System
(HCI)

1) System Vision

The proposed system is an AI-assisted Emergency Department (ED) triage support system designed to assist nurses throughout the patient triage and monitoring process. The implementation consists of a Nurse Priority Dashboard, a detailed Patient Review Screen and a Patient Waiting Portal that allows patients to receive status updates and report changes in their condition while waiting.

The system analyses patient information entered during triage and provides a recommended Emergency Severity Index (ESI) level together with a confidence score and the key clinical factors influencing the recommendation. Patient-reported symptom updates are routed to the nurse dashboard to support timely reassessment.

The system is intended to support clinical decision-making rather than replace it. The triage nurse remains responsible for reviewing recommendations, responding to patient updates, accepting or overriding predictions and making the final clinical decision. The system is designed to integrate into the existing Mercer General Hospital workflow while improving consistency, transparency, communication and patient safety.

2) System Inputs

The system receives the following information during patient assessment.

Input	Source
Patient demographics	Manual entry (future EHR integration)
Presenting complaint	Triage nurse

Blood pressure	Manual entry (future medical device integration)
Heart rate	Manual entry (future medical device integration)
Respiratory rate	Manual entry
Body temperature	Manual entry
Oxygen saturation (SpO ₂)	Manual entry (future medical device integration)
Pain score	Manual entry
Clinical observations	Triage nurse
Patient-reported symptom updates	Patient Waiting Portal
Time since initial triage	System generated
Time since last reassessment	System generated
Nurse override reason	Triage nurse
Reassessment notes	Triage nurse

3) System Processing

After receiving patient information, the system performs the following operations:

- Validates all required input fields.
- Identifies missing or incomplete information.
- Processes the patient data using the trained AI triage model.
- Predicts the recommended Emergency Severity Index (ESI).
- Calculates the confidence level of the prediction.
- Identifies the clinical factors that contributed to the recommendation.
- Receives and validates patient-reported symptom updates.
- Flags patients requiring reassessment.
- Updates the nurse dashboard with new alerts.
- Records reassessment history.

4) System Outputs

The system provides the following outputs to the triage nurse.

5	Output	Purpose
	Recommended ESI Level	Clinical decision support
	Confidence Score	Indicates confidence in prediction
	Colour-coded Priority Indicator	Rapid visual identification of urgency
	Key Contributing Clinical Factors	Improves explainability and transparency
	Nurse Override Option	Maintains human oversight
	Audit Log	Records recommendations and clinical decisions
	Patient deterioration alert	Prompt nurse reassessment
	Patient waiting status	Improve patient communication
	Symptom update acknowledgement	Confirm patient submission
	Reassessment reminder	Support ongoing monitoring

5) Functional Requirements

The system shall:

- Display the predicted ESI classification within **1.5 seconds** of data submission.
- Display the AI confidence score for every prediction.
- Present the key clinical factors influencing the recommendation.
- Allow nurses to accept or override the AI recommendation.
- Require an override reason before the final decision is submitted.
- Record recommendations, overrides and timestamps within an audit log.
- Support manual entry of patient information.
- Display patient information in a single dashboard to minimise navigation during triage.

- Display patient deterioration alerts on the priority dashboard.
- Allow patients to submit symptom updates through the waiting portal.
- Notify nurses when new patient updates are received.
- Record reassessments and patient update history.
- Display estimated waiting status without guaranteeing treatment times.
- Prevent patient portal submissions from automatically changing ESI classification.

6) Non-Functional Requirements

The system shall:

- Remain clearly legible under hospital lighting conditions.
- Support efficient use during busy Emergency Department workflows.
- Minimise cognitive workload through a simple and intuitive interface.
- Continue operating during periods of high patient volume.
- Protect patient confidentiality and privacy.
- Provide clear visual indicators for high-priority patients.
- Maintain high system availability during clinical use.
- Allow manual clinical workflows if AI recommendations become unavailable.
- Support WCAG-informed accessibility principles through readable text and high-contrast colours.
- Maintain secure communication between the patient portal and hospital systems.
- Minimise disruption to existing clinical workflows.

7) Integration Requirements

The system shall:

- Support future integration with the Mercer General Hospital Electronic Health Record (EHR).
- Support future integration with medical devices such as blood pressure monitors, pulse oximeters and thermometers.
- Connect securely to hospital authentication systems.
- Store audit logs securely for governance and quality assurance.
- Allow manual data entry whenever EHR connectivity is unavailable.
- Support future reporting and performance monitoring.
- Support secure QR-code authentication for patient portal access.
- Synchronise patient portal updates with the nurse dashboard.

8) Human Interaction Requirements

The triage nurse shall:

- Review the AI recommendation.
- Review the confidence score and contributing clinical factors.
- Accept or override the recommendation.
- Provide a reason when overriding the AI recommendation.
- Make the final clinical triage decision.
- Review patient deterioration alerts.
- Perform reassessment where required.

The AI system shall:

- Provide recommendations only.
- Never make the final triage decision.
- Clearly indicate that all recommendations are advisory.

The patient shall:

- View a general waiting status.
- Submit changes in symptoms.
- Receive confirmation that updates were submitted.
- Be instructed to seek immediate staff assistance if symptoms become severe.

9) Safety Requirements

The system shall:

- Never automatically assign a patient's final triage category.
- Clearly display that recommendations are intended to support, not replace, clinical judgement.
- Record all override decisions within the audit log.
- Flag incomplete or missing patient information before generating recommendations.
- Support manual clinical decision-making at all times.
- Prioritise patient safety by ensuring that human oversight remains mandatory.
- Patient symptom updates shall never automatically change the assigned ESI level.
- High-priority deterioration alerts shall require nurse acknowledgement.
- Patient portal messages shall clearly state that AI does not replace clinical staff.

10) Summary

The proposed AI-assisted Emergency Department triage support system combines a nurse-facing priority dashboard, detailed patient review interface and patient waiting portal to improve communication, transparency and consistency during emergency department triage. The system supports explainable AI recommendations, structured patient updates, secure audit logging and future integration with hospital information systems while ensuring that nurses remain responsible for every final clinical decision.